

## Initiating Search

November 11, 2025, 9:07 PM

• Search: Selenium dioxide

Filtered By:

Yield: 90-100%, 80-89%, 70-79%

Document Type: Journal

## Search Tasks

| Task                                                | Result Type                                                                                 | View                         |
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| Exported: Returned Reaction Results + Filters (557) |  Reactions | <a href="#">View Results</a> |

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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (83) Suppliers (3)

31-494-CAS-13540885

Steps: 1 Yield: 100%

**Synthesis and anti-protozoal activity of C2-substituted polyazamacrocycles**1.1 Reagents: **Selenium dioxide**

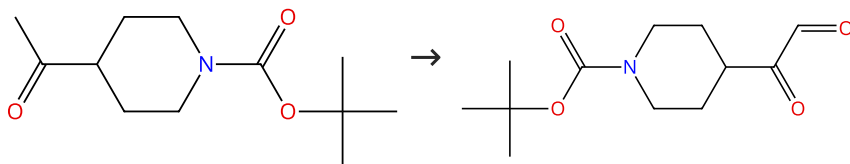
Solvents: 1,4-Dioxane, Water; 18 - 168 h, reflux

By: Reid, Caroline M.; et al

Bioorganic &amp; Medicinal Chemistry Letters (2008), 18(7), 2455-2458.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (83) Suppliers (22)

31-614-CAS-31459024

Steps: 1 Yield: 100%

**Discovery of Hydroxyamidine Derivatives as Highly Potent, Selective Indoleamine-2,3-dioxygenase 1 Inhibitors**1.1 Reagents: **Selenium dioxide**

Solvents: 1,4-Dioxane; 16 h, 80 °C

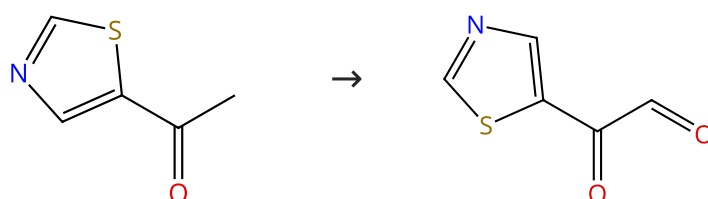
By: Jin, Fangfang; et al

Experimental Protocols

ACS Medicinal Chemistry Letters (2021), 12(2), 195-201.

Scheme 3 (1 Reaction)

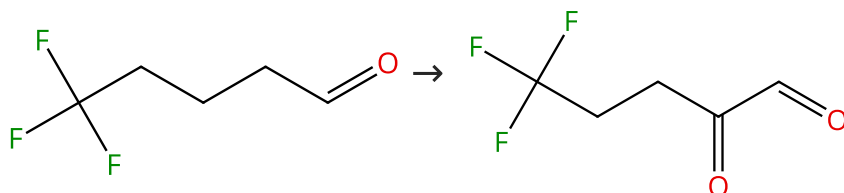
Steps: 1 Yield: 100%

 Suppliers (62) Suppliers (3)

|                                                                            |                      |                                                                                                                     |
|----------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-614-CAS-31458898                                                        | Steps: 1 Yield: 100% | <b>Discovery of Hydroxyamidine Derivatives as Highly Potent, Selective Indoleamine-2,3-dioxygenase 1 Inhibitors</b> |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane; 6 h, 90 °C |                      | By: Jin, Fangfang; et al                                                                                            |
| Experimental Protocols                                                     |                      | ACS Medicinal Chemistry Letters (2021), 12(2), 195-201.                                                             |

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (28)

Suppliers (5)

|                                                                                                       |                      |                                                                                                                      |
|-------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-19290563                                                                                   | Steps: 1 Yield: 100% | <b>Discovery of chiral dihydropyridopyrimidinones as potent, selective and orally bioavailable inhibitors of AKT</b> |
| 1.1 Reagents: Acetic acid, <b>Selenium dioxide</b> , Water<br>Solvents: 1,4-Dioxane; overnight, 90 °C |                      | By: Parthasarathy, Saravanam; et al                                                                                  |
|                                                                                                       |                      | Bioorganic & Medicinal Chemistry Letters (2018), 28(10), 1887-1891.                                                  |

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



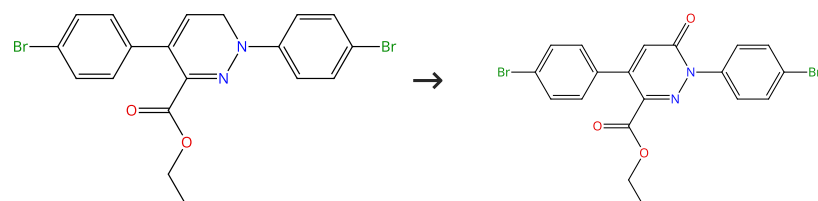
Suppliers (77)

Suppliers (7)

|                                                                                     |                      |                                                                                                  |
|-------------------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------|
| 31-494-CAS-7923651                                                                  | Steps: 1 Yield: 100% | <b>Synthesis and anti-HIV activity evaluation of new phenyl ethyl thiourea (PET) derivatives</b> |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane, Water; 12 h, 100 °C |                      | By: Ettari, Roberta; et al                                                                       |
|                                                                                     |                      | ARKIVOC (Gainesville, FL, United States) (2009), (14), 227-234.                                  |

Scheme 6 (1 Reaction)

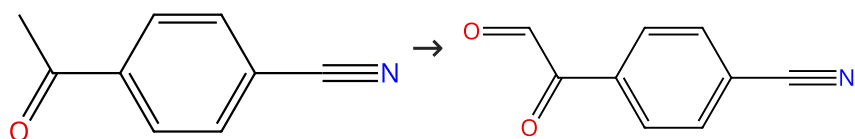
Steps: 1 Yield: 100%



|                                                                             |                      |                                                                                                                                                                                 |
|-----------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-21432386                                                         | Steps: 1 Yield: 100% | <b>Silver(I)-Catalyzed Tandem Sigmatropic Rearrangement/1,3-H Shift/6π Aza-electrocyclization of N-Propargylic Hydrazones: A Mild Synthetic Route to 1,6-Dihydropyridazines</b> |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane; 4 h, 105 °C |                      | By: Ding, Zong-Cang; et al                                                                                                                                                      |
| Experimental Protocols                                                      |                      | Journal of Organic Chemistry (2016), 81(9), 3936-3941.                                                                                                                          |

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (94)

Suppliers (33)

31-494-CAS-1050122

Steps: 1 Yield: 100%

**Synthesis of novel benzamidine- and guanidine-derived polyazamacrocycles: Selective anti-protozoal activity for human African trypanosomiasis**

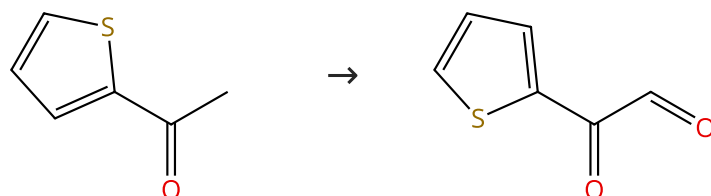
By: Reid, Caroline M.; et al

Bioorganic &amp; Medicinal Chemistry Letters (2008), 18(20), 5399-5401.

1.1 Reagents: **Selenium dioxide**  
Solvents: 1,4-Dioxane, Water; 3 d, reflux

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

Suppliers (34)

31-494-CAS-9277926

Steps: 1 Yield: 100%

**Synthesis and anti-protozoal activity of C2-substituted polyazamacrocycles**

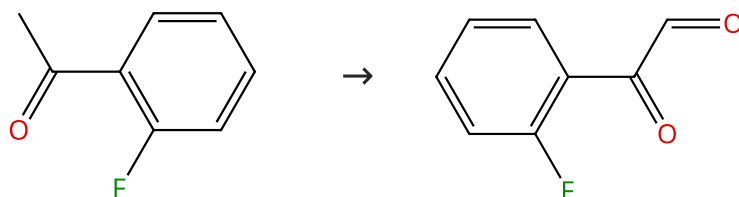
By: Reid, Caroline M.; et al

Bioorganic &amp; Medicinal Chemistry Letters (2008), 18(7), 2455-2458.

1.1 Reagents: **Selenium dioxide**  
Solvents: 1,4-Dioxane, Water; 18 - 168 h, reflux

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (94)

Suppliers (29)

31-494-CAS-11908335

Steps: 1 Yield: 100%

**Synthesis and anti-HIV activity evaluation of new phenyl ethyl thiourea (PET) derivatives**

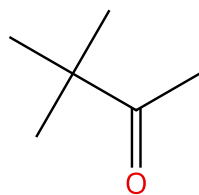
By: Ettari, Roberta; et al

ARKIVOC (Gainesville, FL, United States) (2009), (14), 227-234.

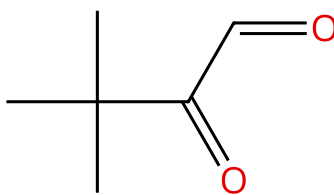
1.1 Reagents: **Selenium dioxide**  
Solvents: 1,4-Dioxane, Water; 12 h, 100 °C

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (39)



Suppliers (40)

31-494-CAS-4729912

Steps: 1 Yield: 100%

**Synthesis and anti-protozoal activity of C2-substituted polyazamacrocycles**1.1 Reagents: **Selenium dioxide**

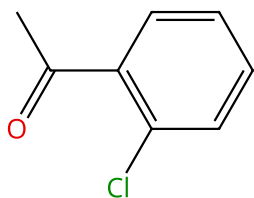
Solvents: 1,4-Dioxane, Water; 18 - 168 h, reflux

By: Reid, Caroline M.; et al

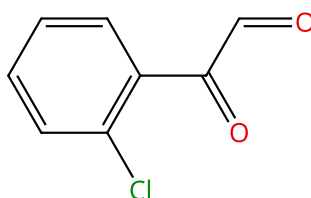
Bioorganic &amp; Medicinal Chemistry Letters (2008), 18(7), 2455-2458.

Scheme 11 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (85)



Suppliers (20)

31-494-CAS-5497548

Steps: 1 Yield: 100%

**Synthesis and anti-HIV activity evaluation of new phenyl ethyl thiourea (PET) derivatives**1.1 Reagents: **Selenium dioxide**

Solvents: 1,4-Dioxane, Water; 12 h, 100 °C

By: Ettari, Roberta; et al

ARKIVOC (Gainesville, FL, United States) (2009), (14), 227-234.

31-494-CAS-359521

Steps: 1 Yield: 100%

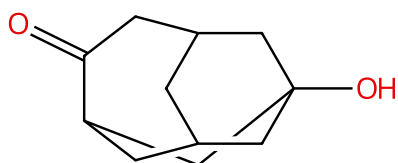
**Synthesis and bronchodilating activity of o-chloro-α-(tert-butylaminomethyl)-benzylalcohol hydrochloride (C-78) and related compounds**1.1 Reagents: **Selenium dioxide**

By: Koshinaka, Eiichi; et al

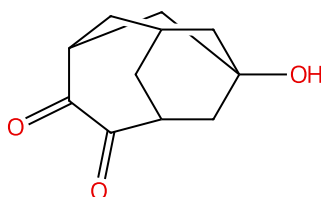
Yakugaku Zasshi (1978), 98(9), 1198-207.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



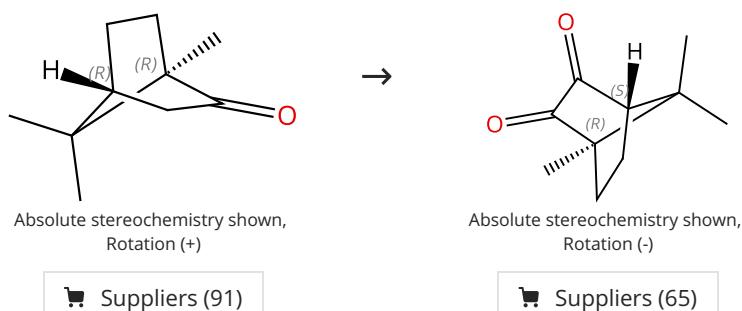
Supplier (1)



|                                                                                |                      |                                                                                             |
|--------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------|
| 31-494-CAS-441652                                                              | Steps: 1 Yield: 100% | <b>Convenient synthesis of novel 1,3,7-trisubstituted bicyclo [3.3.1]nonane derivatives</b> |
| 1.1 Reagents: <a href="#">Selenium dioxide</a><br>Solvents: 1,4-Dioxane, Water |                      | By: Klaic, Lada; et al                                                                      |
| Experimental Protocols                                                         |                      | Synthetic Communications (2002), 32(1), 89-97.                                              |

Scheme 13 (2 Reactions)

Steps: 1 Yield: 99-100%

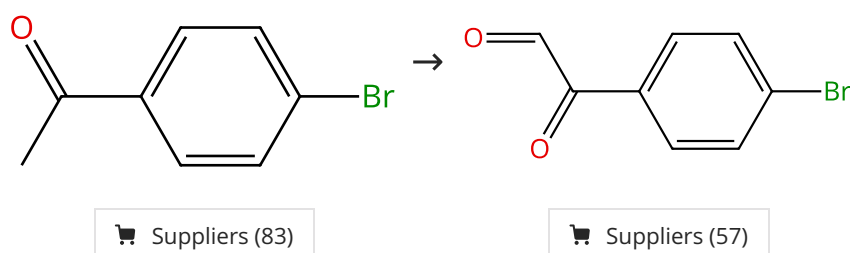


|                                                                              |                      |                                                                                                                                                            |
|------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-15346583                                                          | Steps: 1 Yield: 100% | <b>Chiral Tricyclic Iminolactone Derived from (1R)-(+)-Camphor as a Glycine Equivalent for the Asymmetric Synthesis of <math>\alpha</math>-Amino Acids</b> |
| 1.1 Reagents: <a href="#">Selenium dioxide</a><br>Solvents: Acetic anhydride |                      | By: Xu, Peng-Fei; et al                                                                                                                                    |
| Experimental Protocols                                                       |                      | Journal of Organic Chemistry (2002), 67(7), 2309-2314.                                                                                                     |

|                                                                                                                                      |                     |                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-6076646                                                                                                                   | Steps: 1 Yield: 99% | <b>Enantioselective Addition of Diethylzinc to Aldehydes Catalyzed by Chiral O,N,O-tridentate Phenol Ligands Derived From Camphor</b> |
| 1.1 Reagents: <a href="#">Selenium dioxide</a><br>Solvents: Acetic anhydride; 19 h, rt $\rightarrow$ reflux; reflux $\rightarrow$ rt |                     | By: Lee, Dong-Sheng; et al                                                                                                            |
| 1.2 Reagents: Sodium hydroxide<br>Solvents: Water                                                                                    |                     | Chirality (2016), 28(1), 65-71.                                                                                                       |
| Experimental Protocols                                                                                                               |                     |                                                                                                                                       |

Scheme 14 (1 Reaction)

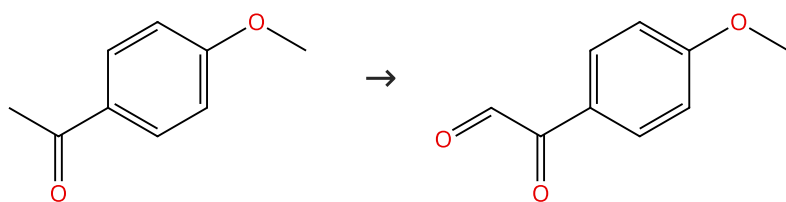
Steps: 1 Yield: 100%



|                                                                                                    |                      |                                                                                   |
|----------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------|
| 31-494-CAS-13248869                                                                                | Steps: 1 Yield: 100% | <b>Synthesis and anti-protozoal activity of C2-substituted polyazamacrocycles</b> |
| 1.1 Reagents: <a href="#">Selenium dioxide</a><br>Solvents: 1,4-Dioxane, Water; 18 - 168 h, reflux |                      | By: Reid, Caroline M.; et al                                                      |
|                                                                                                    |                      | Bioorganic & Medicinal Chemistry Letters (2008), 18(7), 2455-2458.                |

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (97)

Suppliers (46)

31-494-CAS-8997490

Steps: 1 Yield: 100%

**Synthesis and anti-protozoal activity of C2-substituted polyazamacrocycles**

1.1 Reagents: Selenium dioxide

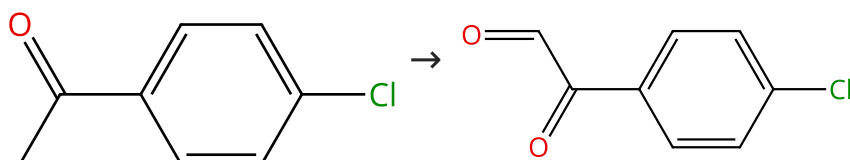
Solvents: 1,4-Dioxane, Water; 18 - 168 h, reflux

By: Reid, Caroline M.; et al

Bioorganic &amp; Medicinal Chemistry Letters (2008), 18(7), 2455-2458.

Scheme 16 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (67)

Suppliers (32)

31-494-CAS-9780346

Steps: 1 Yield: 100%

**Synthesis and anti-HIV activity evaluation of new phenyl ethyl thiourea (PET) derivatives**

1.1 Reagents: Selenium dioxide

Solvents: 1,4-Dioxane, Water; 12 h, 100 °C

By: Ettari, Roberta; et al

ARKIVOC (Gainesville, FL, United States) (2009), (14), 227-234.

31-494-CAS-11121887

Steps: 1 Yield: 100%

**Synthesis and anti-protozoal activity of C2-substituted polyazamacrocycles**

1.1 Reagents: Selenium dioxide

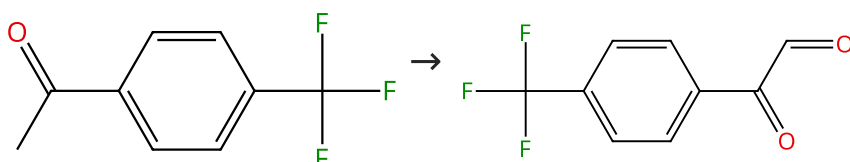
Solvents: 1,4-Dioxane, Water; 18 - 168 h, reflux

By: Reid, Caroline M.; et al

Bioorganic &amp; Medicinal Chemistry Letters (2008), 18(7), 2455-2458.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



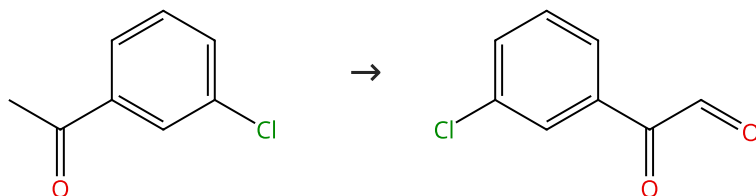
Suppliers (85)

Suppliers (41)

|                                                                                           |                                                                                                        |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>31-494-CAS-15378701</b> Steps: 1 Yield: 100%                                           | <b>Synthesis and anti-protozoal activity of C2-substituted polyazamacrocycles</b>                      |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane, Water; 18 - 168 h, reflux | By: Reid, Caroline M.; et al<br><br>Bioorganic & Medicinal Chemistry Letters (2008), 18(7), 2455-2458. |

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



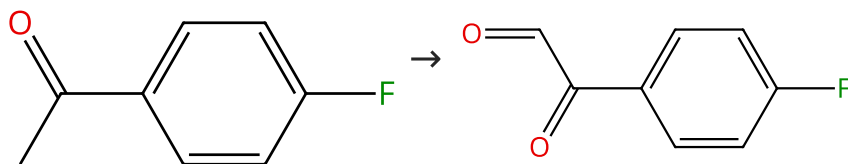
Suppliers (58)

Suppliers (26)

|                                                                                     |                                                                                                   |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>31-494-CAS-7649428</b> Steps: 1 Yield: 100%                                      | <b>Synthesis and anti-HIV activity evaluation of new phenyl ethyl thiourea (PET) derivatives</b>  |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane, Water; 12 h, 100 °C | By: Ettari, Roberta; et al<br><br>ARKIVOC (Gainesville, FL, United States) (2009), (14), 227-234. |

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



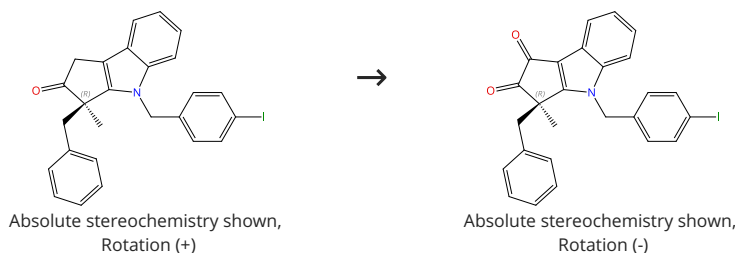
Suppliers (93)

Suppliers (48)

|                                                                                     |                                                                                                   |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>31-494-CAS-14029590</b> Steps: 1 Yield: 100%                                     | <b>Synthesis and anti-HIV activity evaluation of new phenyl ethyl thiourea (PET) derivatives</b>  |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane, Water; 12 h, 100 °C | By: Ettari, Roberta; et al<br><br>ARKIVOC (Gainesville, FL, United States) (2009), (14), 227-234. |

Scheme 20 (1 Reaction)

Steps: 1 Yield: 99%


 Absolute stereochemistry shown,  
 Rotation (+)

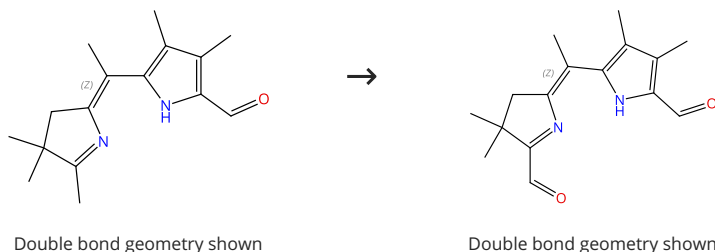
 Absolute stereochemistry shown,  
 Rotation (-)



|                                                                                     |                     |                                                                                                          |
|-------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------|
| 31-491-CAS-23436347                                                                 | Steps: 1 Yield: 99% | <b>Catalytic Enantioselective Desymmetrizing Fischer Indolization through Dynamic Kinetic Resolution</b> |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane, Water; 24 h, 110 °C |                     | By: Ghosh, Biki; et al                                                                                   |
| Experimental Protocols                                                              |                     | Angewandte Chemie, International Edition (2021), 60(16), 9086-9092.                                      |

Scheme 21 (1 Reaction)

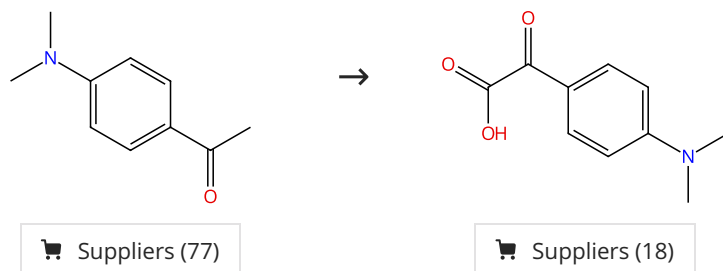
Steps: 1 Yield: 99%



|                                                                |                     |                                        |
|----------------------------------------------------------------|---------------------|----------------------------------------|
| 31-494-CAS-3624390                                             | Steps: 1 Yield: 99% | <b>A New Synthesis of Chlorins</b>     |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane |                     | By: Jacobi, Peter A.; et al            |
| 1.2 Reagents: Sodium bicarbonate<br>Solvents: Dichloromethane  |                     | Organic Letters (2001), 3(6), 831-834. |
| Experimental Protocols                                         |                     |                                        |

Scheme 22 (1 Reaction)

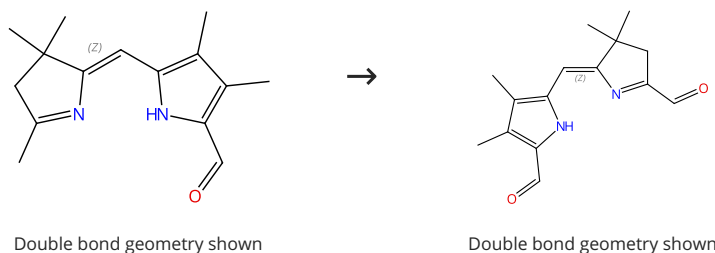
Steps: 1 Yield: 99%



|                                                 |                     |                                                                                                                     |
|-------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-14610849                             | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective Haemophilus influenzae antibacterial agents</b> |
| 1.1 Reagents: Pyridine, <b>Selenium dioxide</b> |                     | By: Beebe, Xenia; et al                                                                                             |
|                                                 |                     | Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                                                 |

Scheme 23 (1 Reaction)

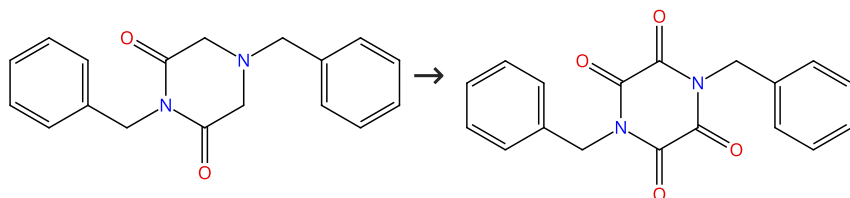
Steps: 1 Yield: 99%



|                                                                |                     |                                        |
|----------------------------------------------------------------|---------------------|----------------------------------------|
| 31-494-CAS-2947931                                             | Steps: 1 Yield: 99% | <b>A New Synthesis of Chlorins</b>     |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane |                     | By: Jacobi, Peter A.; et al            |
| 1.2 Reagents: Sodium bicarbonate<br>Solvents: Dichloromethane  |                     | Organic Letters (2001), 3(6), 831-834. |
| Experimental Protocols                                         |                     |                                        |

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



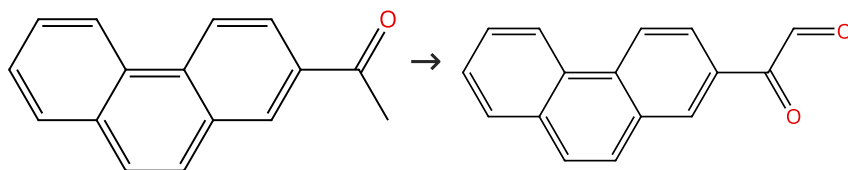
Suppliers (7)

Suppliers (3)

|                                       |                     |                                                                  |
|---------------------------------------|---------------------|------------------------------------------------------------------|
| 31-494-CAS-11688182                   | Steps: 1 Yield: 99% | <b>Oxidation of 2,6-piperazinediones</b>                         |
| 1.1 Reagents: <b>Selenium dioxide</b> |                     | By: Tanaka, Tatsuo; et al                                        |
|                                       |                     | Bulletin of the Chemical Society of Japan (1977), 50(7), 1821-3. |

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%



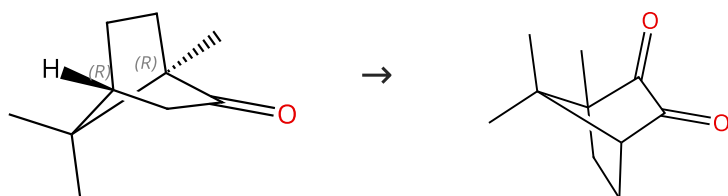
Suppliers (57)

Suppliers (2)

|                                                                                           |                     |                                                                                   |
|-------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------|
| 31-494-CAS-2276655                                                                        | Steps: 1 Yield: 99% | <b>Synthesis and anti-protozoal activity of C2-substituted polyazamacrocycles</b> |
| 1.1 Reagents: <b>Selenium dioxide</b><br>Solvents: 1,4-Dioxane, Water; 18 - 168 h, reflux |                     | By: Reid, Caroline M.; et al                                                      |
|                                                                                           |                     | Bioorganic & Medicinal Chemistry Letters (2008), 18(7), 2455-2458.                |

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown,  
Rotation (+)

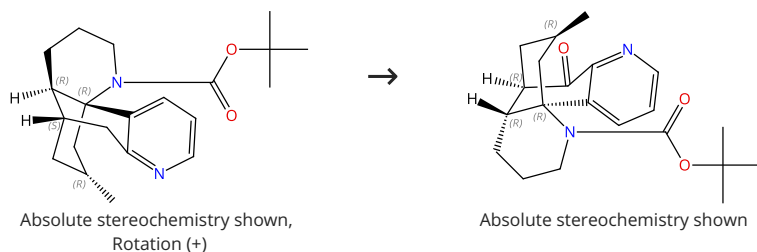
Suppliers (91)

Suppliers (100)

|                                                                                             |                     |                                                                                                                        |
|---------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-5389849                                                                          | Steps: 1 Yield: 99% | Copper(I) bromide-mediated coupling of imidazoles. Trapping and x-ray crystal structures of the reaction intermediates |
| 1.1 Reagents: Acetic anhydride, Selenium dioxide<br>Solvents: Acetic anhydride; 5 h, reflux |                     | By: Huang, Xue-Feng; et al                                                                                             |
| Experimental Protocols                                                                      |                     | Crystal Growth & Design (2008), 8(6), 1795-1797.                                                                       |

Scheme 27 (1 Reaction)

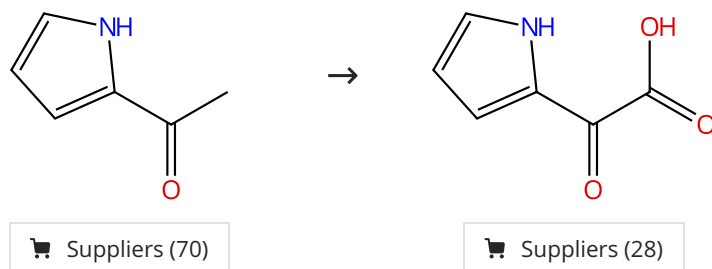
Steps: 1 Yield: 99%



|                                                                      |                     |                                                                                             |
|----------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------|
| 31-491-CAS-8995007                                                   | Steps: 1 Yield: 99% | Synthetic Studies on Pseudo-Dimeric Lycopodium Alkaloids: Total Synthesis of Complanadine B |
| 1.1 Reagents: Selenium dioxide<br>Solvents: 1,4-Dioxane; 2 h, 150 °C |                     | By: Newton, James N.; et al                                                                 |
| Experimental Protocols                                               |                     | Angewandte Chemie, International Edition (2013), 52(6), 1726-1730.                          |

Scheme 28 (1 Reaction)

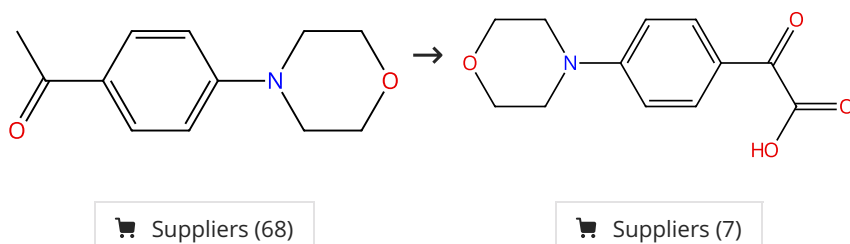
Steps: 1 Yield: 99%



|                                          |                     |                                                                                                              |
|------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-5032176                       | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective Haemophilus influenzae antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al                                                                                      |
|                                          |                     | Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                                          |

Scheme 29 (1 Reaction)

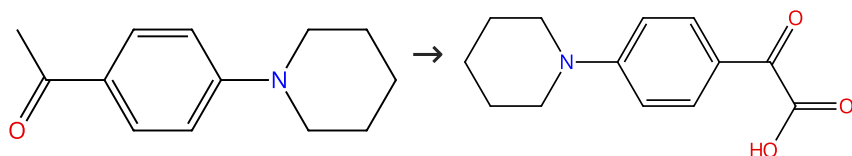
Steps: 1 Yield: 99%



|                                                 |                     |                                                                                                                            |
|-------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-14010759                             | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, <b>Selenium dioxide</b> |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 30 (1 Reaction)

Steps: 1 Yield: 99%



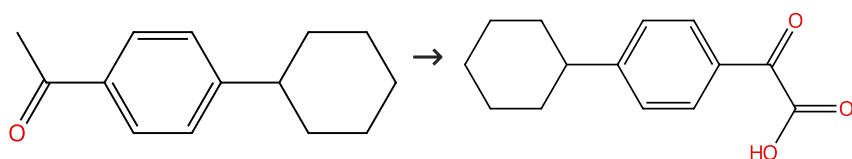
Suppliers (59)

Suppliers (8)

|                                                 |                     |                                                                                                                            |
|-------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-616617                               | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, <b>Selenium dioxide</b> |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 31 (1 Reaction)

Steps: 1 Yield: 99%



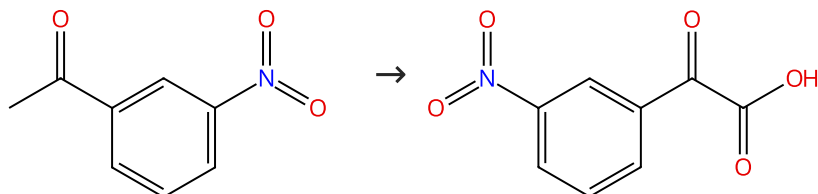
Suppliers (70)

Suppliers (16)

|                                                 |                     |                                                                                                                            |
|-------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-11879598                             | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, <b>Selenium dioxide</b> |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



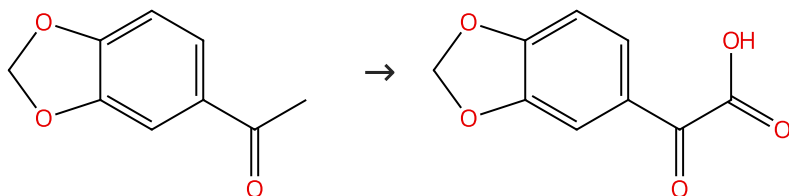
Suppliers (83)

Suppliers (10)

|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-7944971                       | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



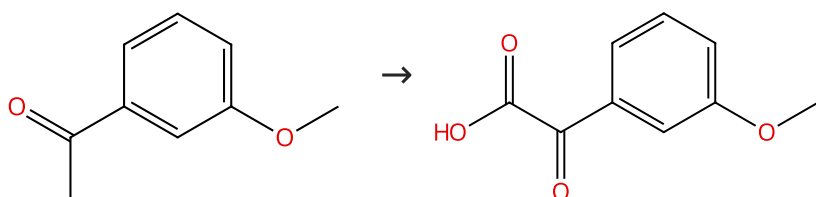
Suppliers (74)

Suppliers (53)

|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-949190                        | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 34 (1 Reaction)

Steps: 1 Yield: 99%



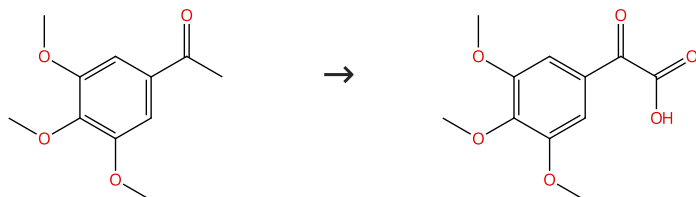
Suppliers (89)

Suppliers (44)

|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-3338078                       | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%



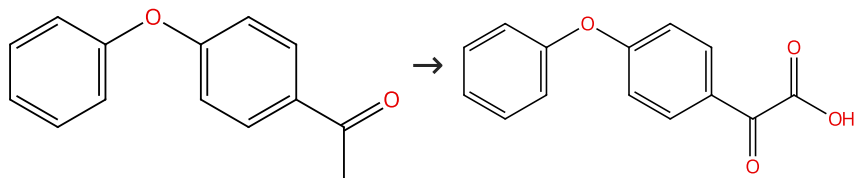
Suppliers (72)

Suppliers (55)

|                                                 |                     |                                                                                                                            |
|-------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-12734286                             | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, <b>Selenium dioxide</b> |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



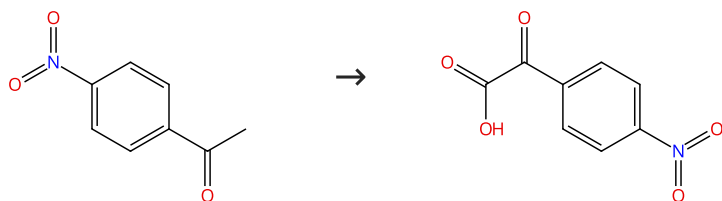
Suppliers (85)

Suppliers (37)

|                                                 |                     |                                                                                                                            |
|-------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-9745458                              | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, <b>Selenium dioxide</b> |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 37 (1 Reaction)

Steps: 1 Yield: 99%



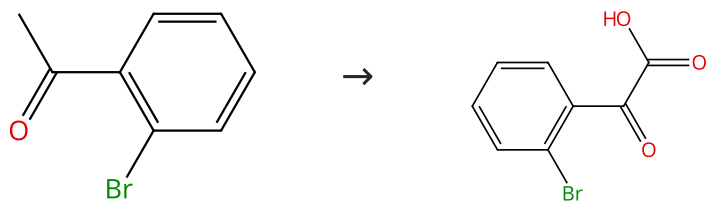
Suppliers (78)

Suppliers (59)

|                                                 |                     |                                                                                                                            |
|-------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-5469703                              | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, <b>Selenium dioxide</b> |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 38 (1 Reaction)

Steps: 1 Yield: 99%



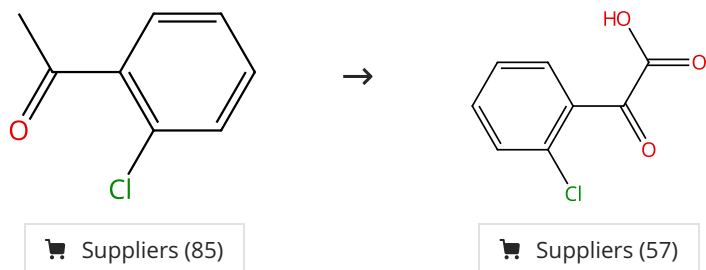
Suppliers (88)

Suppliers (50)

|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-8447658                       | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 39 (1 Reaction)

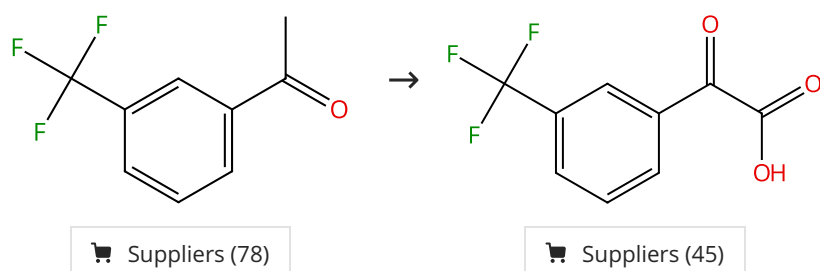
Steps: 1 Yield: 99%



|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-10584158                      | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 40 (1 Reaction)

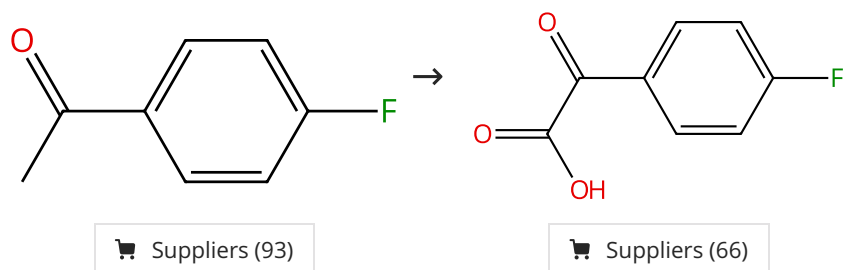
Steps: 1 Yield: 99%



|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-3157336                       | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 41 (1 Reaction)

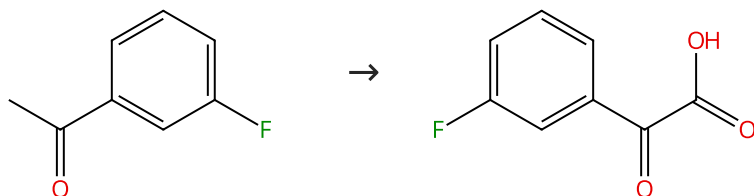
Steps: 1 Yield: 99%



|                                          |                     |                                                                                                                            |
|------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-12202242                      | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 42 (1 Reaction)

Steps: 1 Yield: 99%



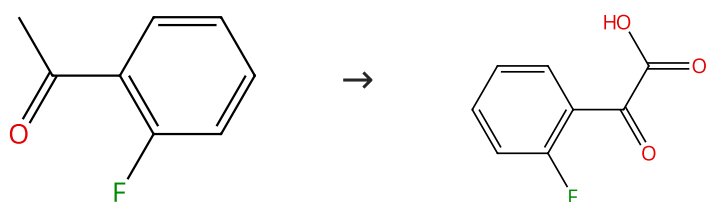
Suppliers (82)

Suppliers (55)

|                                          |                     |                                                                                                                            |
|------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-10073400                      | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%



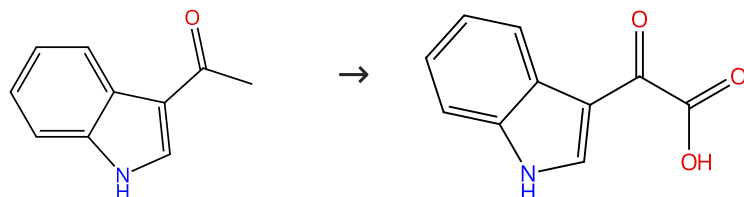
Suppliers (94)

Suppliers (56)

|                                          |                     |                                                                                                                            |
|------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-7611274                       | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b> |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                             |

Scheme 44 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (95)

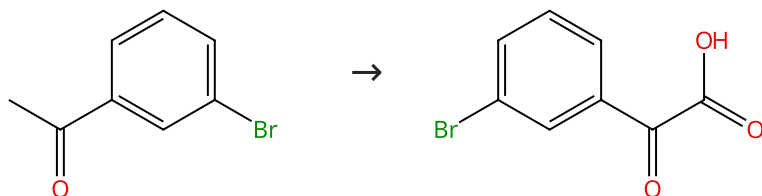
Suppliers (67)



|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-7178192                       | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 45 (1 Reaction)

Steps: 1 Yield: 99%



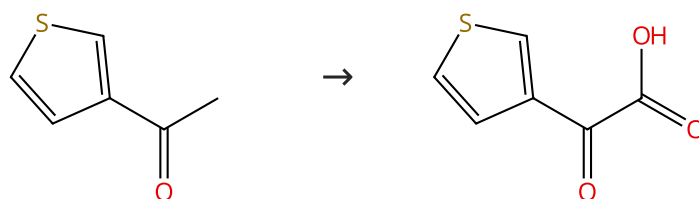
Suppliers (86)

Suppliers (56)

|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-14941556                      | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 46 (1 Reaction)

Steps: 1 Yield: 99%



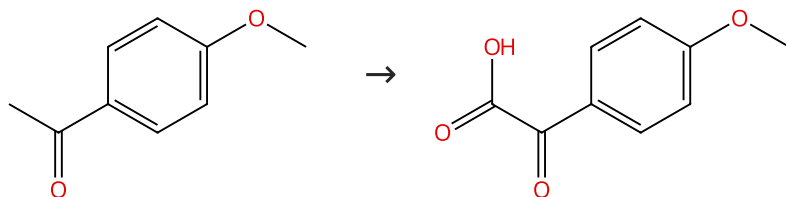
Suppliers (88)

Suppliers (19)

|                                          |                     |                                                                                                                     |
|------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-2921776                       | Steps: 1 Yield: 99% | Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     | By: Beebe, Xenia; et al<br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136.                      |

Scheme 47 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (97)

Suppliers (55)

|                                          |                     |                                                                                                                                                                                                                                      |
|------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-494-CAS-1223118                       | Steps: 1 Yield: 99% | <b>Synthesis and SAR evaluation of oxadiazolopyrazines as selective <i>Haemophilus influenzae</i> antibacterial agents</b><br><br>By: Beebe, Xenia; et al<br><br>Bioorganic & Medicinal Chemistry Letters (2003), 13(19), 3133-3136. |
| 1.1 Reagents: Pyridine, Selenium dioxide |                     |                                                                                                                                                                                                                                      |